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Final Project

Pokémon Game

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1. Introduction and System description

We want for this project to be a kind of Pokédex in base of a Pokémon collection. We will use Java and we want to include graphic interface. This application will allow to the user visualise, modify and manage the information of the different pokémons and create new ones saved in a text file, those pokémons will be managed by user using simple login.

In addition to basic management, the system includes simple mechanical combat between pokémons. Each pokémon will have attributes like life, stamina and a limit attack set. This system will incorporate a module of user management. The login information will be saved on a text file.

This design has been planned in a modular way, allowing future expansions such as improvements in combat system, inclusion of more attributes or

2. Requirement Specification

FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT	DESCRIPTION
FR1	Users sign up	System will allow create new users with a username and password.
FR2	Log in	User will be able to log in using their credentials.
FR3	Pokémon launch	System will upload Pokémons from a text file.
FR4	Show Pokémons	User will be able to watch the list of available Pokémons.
FR5	Modify Pokémons name	User will be able to change the name of a Pokémon.
FR6	Evolve Pokémon	User will be able to evolve a Pokémon if it meets conditions.
FR7	Train Pokémon	User will be able to power up their basic statistics.
FR8	Start combat	User will be able to start an attack choosing first one Pokémon.
FR9	Combat system	Combat will be based on life, stamina and attacks.
FR10	Attacks section	User will be able to choose at least two attacks.
FR11	Save data	Changes made will be saved on a text file.
FR12	Read users	System will read users from a text file.
FR13	Pokémon Launch	

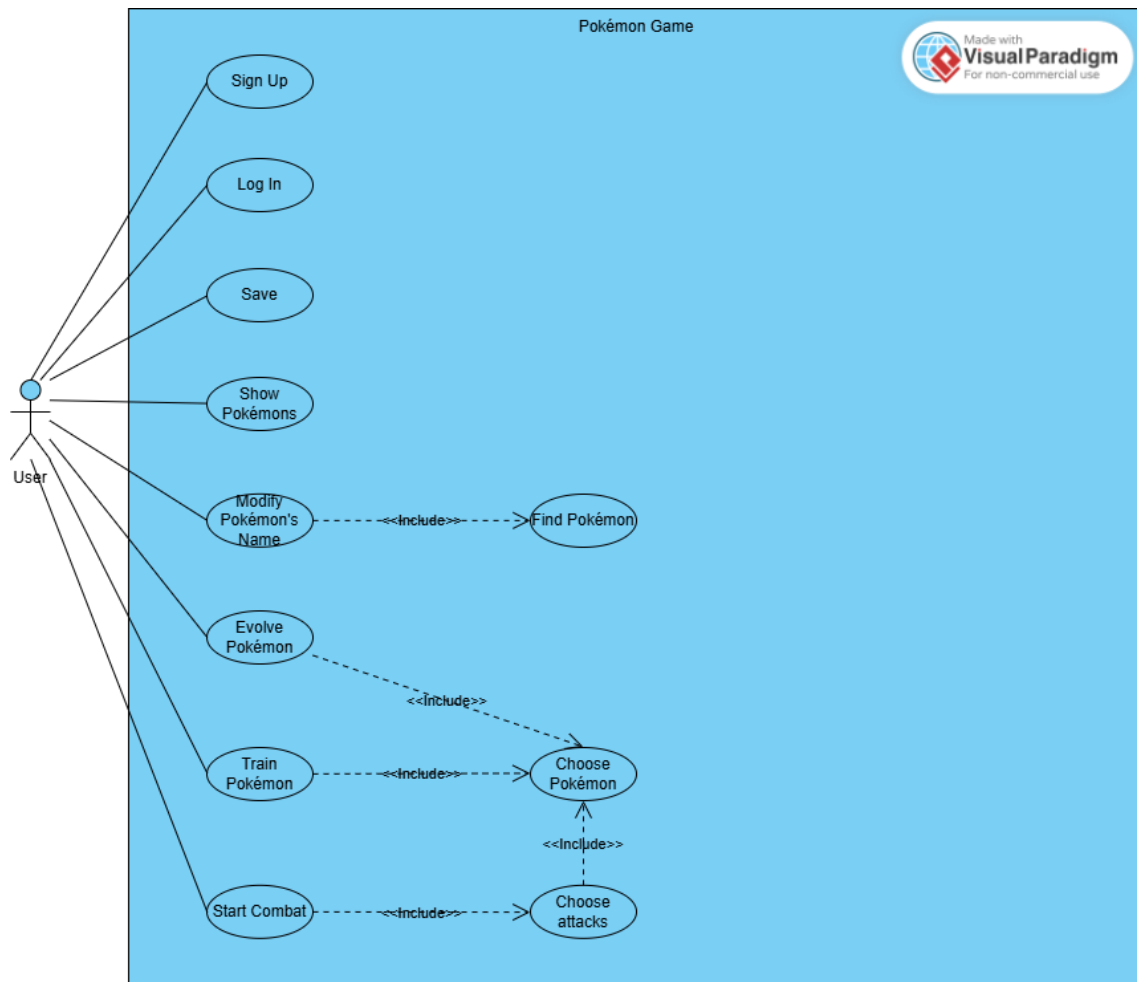
NON-FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT	DESCRIPTION
NFR1	Utility	Graphic interface must be simple and intuitive.
NFR2	Performance	App must answer quickly basic actions.
NFR3	Persistence	Data must be maintained between runs.
NFR4	Scalability	System must allow add more functionalities in the future.
NFR5	Maintainability	Code must be structured.
NFR6	Basic security	Password will be stored simply.
NFR7	Compatibility	App must work in any system with Java.

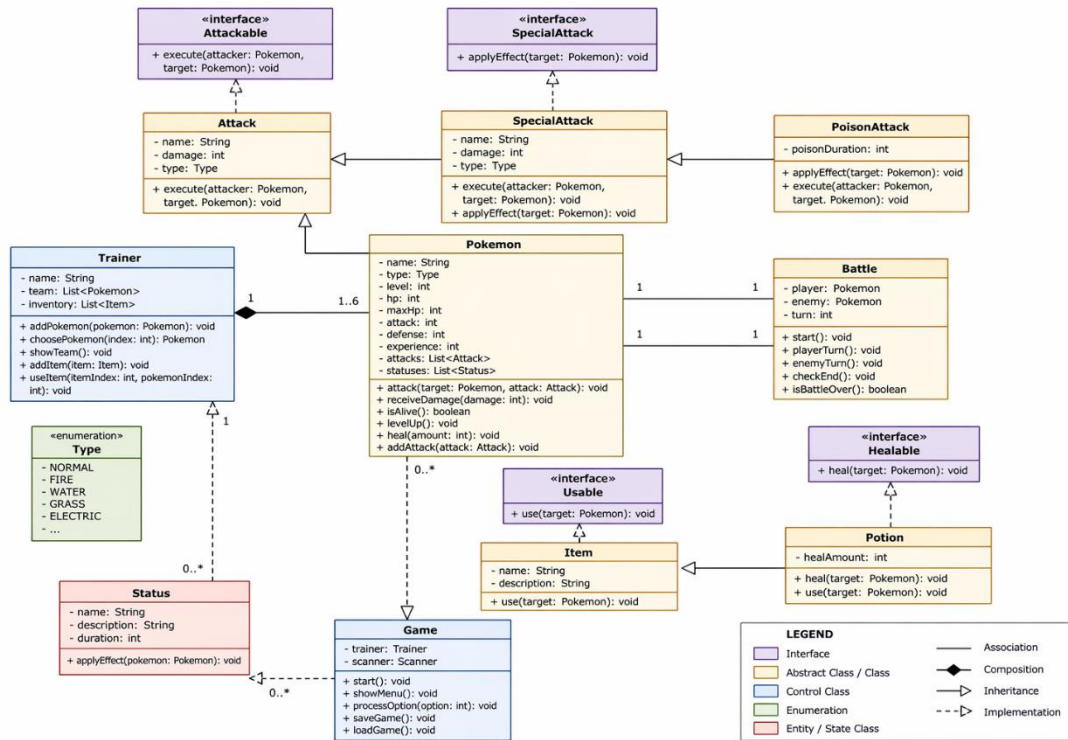
SYSTEM REQUIREMENTS

ID	REQUIREMENT	DESCRIPTION
SR1	Data	Data must be saved and updated always.
SR2	Complexity	App must be adapted to our capabilities.
SR3	Interface	The interface must be basic.
SR4	Files	All files must be text files.

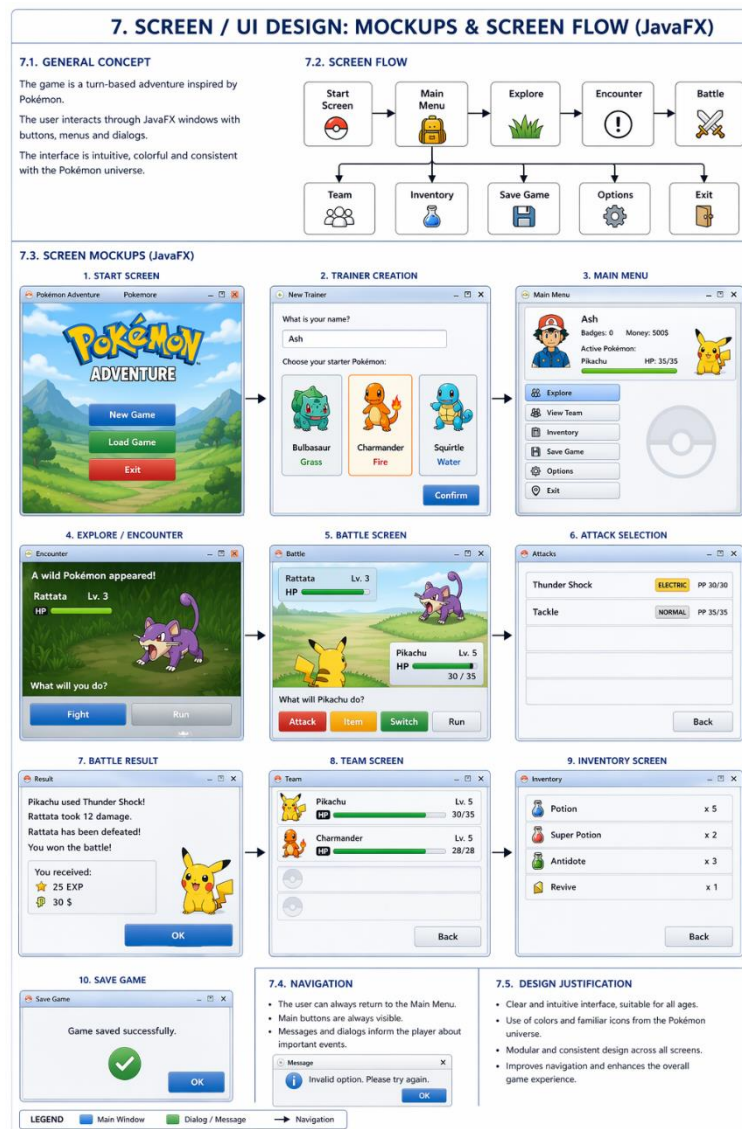
3. Use Case Diagram



4. Class Diagram



5. Screen / UI Design



6. Expected Milestone Schedule

Week 1 – Analysis & Planning

Objectives:

- Define the project idea (Pokémon-style game)
- Identify system requirements
- Plan overall structure

Tasks:

- Write **Introduction & System Description**
- Define **functional requirements** (battle, team, inventory...)
- Define **non-functional requirements** (usability, performance)
- Create **Use Case Diagram**
- Decide main features (scope control)

Deliverables:

- Requirements specification completed
- Use case diagram finished

Week 2 – Introduction to code

Objectives:

- Create classes

Tasks:

- Create the structure of the classes
- Fill the classes, enum and interfaces
- Create first screen

Deliverables:

- GitHub code

Week 3 – Adding screens and logic

Objectives:

- Create more screens
- Make the log in and sign-up work

Tasks:

- Create the screen of log in and sign-up
- Make it save the user info
- Make it show in the main screen info from the user's file

Deliverables:

- GitHub code

Week 4 – Adding screens and logic

Objectives:

- Create Team-view screen

Tasks:

- Create the team-view screen
- Make it the team-view screen be able to read the information of the user and their pokémons and show it
- Fix some issues from log in and sign-up logic

Deliverables:

- GitHub code

Week 5 – Adding screens and logic

Objectives:

- Create create-team logic and screen
- Create options-screen

Tasks:

- Fill the functions of attack class
- Add at least one option in options screen
- Add create-team

Deliverables:

- GitHub code

Week 6 – Adding screens and logic

Objectives:

- Complete create-team logic and screen
- Check code

Tasks:

- Add comments and generate JavaDoc
- Add testing
- Check functionality of the code

Deliverables:

- GitHub code